Introductory activity

Activity 1 · Measuring one litre with two cans 35 min. We have a 9-litre can, a 7-litre can, both without markings, and a water supply. We may fill a can, empty it, or pour one into the other until one is empty or the other is full.

1. Show that one can get exactly 2 litres in the large can.
2. If we pour in u times 9 litres altogether and throw away v times 7 litres, the quantity left is $9u - 7v$. Find natural numbers u and v such that $9u - 7v = 1$. Deduce how to get 1 litre.
3. The cans are replaced by cans of 6 and 9 litres. Show that every quantity of the form $6u + 9v$, with u and v integers, is a multiple of 3. Can one get 1 litre?
4. With cans of a and b litres, which whole quantities does it seem possible to get? State a conjecture involving the greatest common divisor of a and b.

Observing is not proving: the properties conjectured here are established in the course.